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Interrogation of FBG-based strain sensors by means of laser radio-frequency modulation techniques

G Gagliardi, M Salza, P Ferraro and P De Natale

Istituto Nazionale di Ottica Applicata and European Laboratory for Non-Linear Spectroscopy, via Campi Flegrei 34, I-80078 Pozzuoli (Naples), Italy

E-mail: gagliardi@inoa.it

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Abstract
We report on novel, highly-sensitive methods for interrogation of fibre Bragg gratings (FBGs) as well as high-finesse fibre resonators. Basically, the strain detection technique relies on radio-frequency modulation of a telecom distributed-feedback diode laser with phase-sensitive detection of the sensor-reflected signals. In a first set-up, the optical power from a fibre grating is demodulated at multiples of the sideband frequency and a dispersive signal, which monitors thermal and mechanical stress on the FBG, is generated. A fast Fourier transform analysis of this signal revealed the possibility of detecting dynamic strains up to 20 kHz, this limit being set only by the bandwidth of the test device. Minimum detectable strain levels below 200 nε Hz$^{-1/2}$, in the quasi-static domain (0.5–2 Hz), and between 1 and 4 nε Hz$^{-1/2}$ in the 0.4–1 kHz range, were achieved. A different approach is based on an in-fibre Fabry–Perot cavity, made of an FBG pair with very high peak reflectivity (>99%). In this scheme, the diode laser was actively frequency-locked to the FBG cavity, using the Pound–Drever–Hall technique. The resulting error signal was used as a monitor of the strain suffered by the intra-cavity fibre. We demonstrated that a sensitivity gain of at least one order of magnitude could be obtained by this system in a very compact design. Analysis and quantification of the main limiting factors were also carried out in both cases.

Keywords: fibre Bragg gratings, strain measurements, diode laser, frequency modulation, Fabry–Perot cavity

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Optical fibre sensors, such as fibre Bragg gratings (FBGs), have found several applications in different fields owing to their capability of real-time, *in situ*, sensitive strain and temperature measurements, with the advantage of immunity to electromagnetic interference and potential low cost [1, 2]. In addition, the ability to be directly embedded into materials makes them very attractive tools for creating smart monitoring networks for different environments. A number of interrogation methods have been developed so far, using broadband radiation sources combined with different kinds of optical spectrum analysers. In particular, passive etalons and Fabry–Perot devices as well as active filters, such as integrated acousto-optic tunable filters (IAOTFs), have been extensively used in the area of optical sensing, and the possibility of detecting static and dynamic wavelength shifts of FBGs has been widely shown with different interrogation schemes [3–6]. This choice makes feasible the multiple interrogation of several FBGs written in different fibre optic cables for multiple-point sensing with typical sensitivities in the microstrain range.

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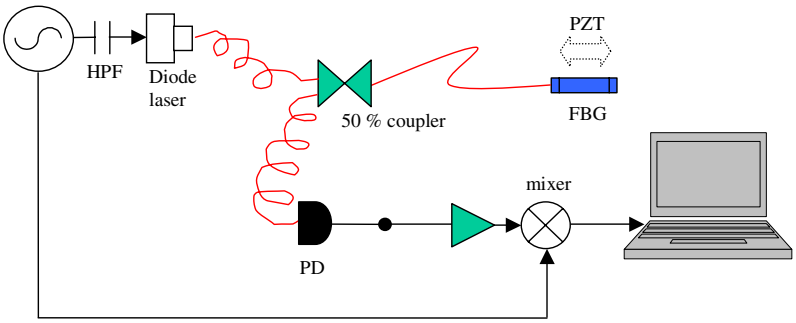

Figure 1. Sketch of the experimental set-up. HPF stands for high-pass filter, PD for photodetector and PZT for piezo-electric transducer.

However, more demanding sensitivity limits may be necessary to monitor special structures, such as high-quality telescopes, long interferometers and precision tools. In addition, the possibility of performing both static and dynamic measurements can be of great relevance in several areas of research, such as geophysics and gravitational wave detection [7]. Impressive performances in FBG-based strain monitoring were obtained, for instance, by means of sophisticated techniques relying on scanning Michelson interferometers that provided accurate wavelength measurements of the grating-reflected light [5]. In this respect, a combination of laser-based detection techniques with FBGs proved to be competitive for high-sensitivity strain measurements [8, 9], also enabling static and dynamic deformations at the nε level [10] to be tracked. Quite recently, new approaches using FBG-based in-fibre resonant structures have been proposed for sensing of strain, pressure and refractive index [11–14].

Here we report on the implementation of two interrogation methods, based on a frequency-modulated distributed-feedback (DFB) diode laser, emitting at 1560 nm, employed either to measure wavelength shifts of an FBG sensor or to track the response of a high-reflectivity FBG-based fibre resonator under applied mechanical stress. The first set-up, whose performance sensitivity has been reported in a recently published paper [15], proved able to detect very small strains with an extremely fast response in a wide frequency range within the acoustic domain. At the same time, a high-finesse cavity-enhanced method, which relies on laser frequency locking onto a cavity longitudinal mode in the Pound–Drever–Hall (PDH) scheme [16], allowed even lower detection limits to be reached in dynamic strain detection. In both cases, our main concern was to achieve a quantitative understanding of every single aspect influencing the performance of the system, including ambient, thermal and laser-induced noise.

2. Experimental methods

The first set-up is depicted in figure 1. A fibre-pigtailed DFB diode laser, emitting at 1560.6 nm, with a maximum power of 20 mW, is driven by a Peltier-based PID controller and a low-noise power source for temperature stabilization and current supply. The current source is also equipped with a 1 MHz-bandwidth modulation input. The laser can be connected to a 50% reflectivity FBG element (AOS GmbH) via a 3 dB coupler that collects the radiation reflected by the grating. The centre wavelength is 1560.78 nm with a 0.07 nm bandwidth. For testing purposes, the FBG is glued onto a piezo-electric actuator with a wide electrical bandwidth (DC to 40 kHz). Hence, static and dynamic longitudinal strain is applied to the fibre grating in a controlled manner. The measurement principle relies on heterodyne detection of the FBG-reflected radiation when the source is tuned into resonance with its spectrum [15].

The laser frequency is modulated at $f = 2.2$ GHz, applying a synthesized radio-frequency signal (R&S model SML03) to the laser current, thus generating symmetric sidebands at a distance f from the carrier frequency. The radiation coming from the grating is detected by an InGaAs PIN photodiode, with a bandwidth of 5 GHz (Thorlabs model SIR5-FC). Demodulation at frequency f is performed by a double-balanced mixer (DBM), while proper phase adjustments are accomplished by a variable delay on the local oscillator (LO). All signals can be continuously acquired through a data acquisition interface by a LabView routine running on a laptop computer.

The high-frequency modulation allows for a strong reduction in noise contributions coming from the laser as well as from detection electronics. The DBM produces a DC voltage output proportional to the superposition of the beat signals of the carrier with the two sidebands, which is extremely sensitive to phase and power differences between them [17]. As a consequence, any mechanical or thermal stress acting on the sensing element shifts the Bragg peak with respect to the laser, thus causing a variation of the resulting signal amplitude. The voltage-to-frequency transfer function can be retrieved by means of simultaneous acquisition of the spectrum of a Fabry–Perot interferometer, using the RF sidebands as a frequency marker. In this way, strain calibration of the DBM output can be readily done if the FBG gauge factor is known. The DC voltage from the mixer represents an ideal monitor of static and dynamic strain within its linear range around the Bragg peak (about 100 $\mu\varepsilon$).

A sketch of the second experimental apparatus is in figure 2. The 1560 nm diode laser is fibre-coupled to an array made of two identical FBGs with peak reflectivity exceeding 99.5%, at a relative distance of about 13 cm. An additional optical isolation stage is used to prevent any disturbance to the laser due to optical feedback. The FBGs were manufactured by AOS GmbH (Dresden) using laser inscription into a hydrogen-loaded single-mode photosensitive fibre. They were both protected by a small aluminium cylinder

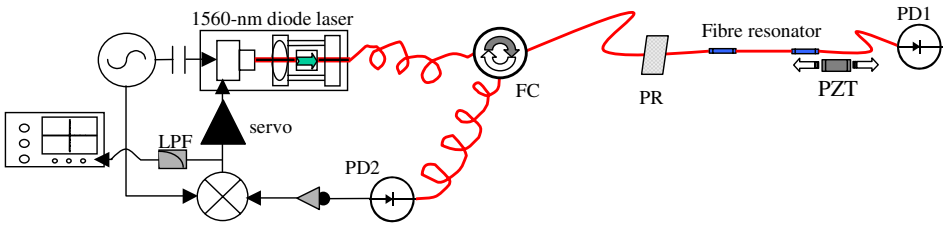

Figure 2. Cavity-based experimental set-up. PD stands for photodiodes, LPF for low-pass filter, FC for fibre circulator and PR for polarization rotator. The loop error signal can be visualized on a digital fast Fourier transform (FFT) analyser with a variable bandwidth while it is sent to the servo input for laser frequency corrections.

in order to keep their Bragg centre wavelengths stable. As the centre wavelengths of the FBGs are substantially coincident, within their spectral width, this structure forms a high-finesse Fabry–Perot resonator with a finesse >500 around 1560 nm and a free-spectral-range (FSR) of approximately 800 MHz. The latter value is in good agreement with predictions of a theoretical model that relates the mode spacing of FBG Fabry–Pérot structures to an equivalent FBG distance, accounting for the actual penetration depth of light propagating into the gratings [18].

A totally different interrogation scheme is adopted in this case. The laser is tightly frequency-locked to the cavity modes by means of an optoelectronic loop, based on the PDH method [16]. The transmission and reflection from the resonator are detected by two pigtailed InGaAs PIN photodiodes. A fibre circulator ensures that the incident and reflected beams are physically separated, while a $\lambda/4 - \lambda/2$ rotation stage controls the polarization state of the incident beam. The laser is frequency-modulated at about 11 MHz, creating sidebands at this frequency distance from the carrier. Since the leakage field back towards the source is basically in antiphase with the input field, the cancellation of these two fields leads to a net reflection coefficient with a strongly frequency-dependent phase shift around the resonance. Hence, the cavity-reflected beam is again phase-sensitive, demodulated by a double-balanced mixer to generate an even-symmetric dispersive-like signal that contains information on attenuation and dispersion differences experienced by the sidebands. The cavity transmission maximum exactly corresponds to vanishing of this signal, which is used as an error signal of a proportional-integrator (PI) feedback loop for frequency stabilization of the laser to the cavity. In this way, the laser follows any frequency shift of the resonance eventually caused by changes in its optical pathlength. Indeed, the thermal and mechanical conditions of the intra-cavity fibre strongly influence the position of its resonances, hence the error delivers information on the strain and temperature variations of the fibre. For the same reason, the sensor was equipped with passive thermal isolation to maintain its temperature stability during measurements. Length variations can be imposed on the intra-cavity fibre by a 150 V piezo-electric transducer (PZT) acting between the ends of the fibre mount. Its response in terms of strain was accurately calibrated over a wide acoustic-frequency interval via cavity-length modulation using the 11 MHz sidebands as a reference, thus accounting for mechanical coupling and resonances of the PZT with the fibre mount.

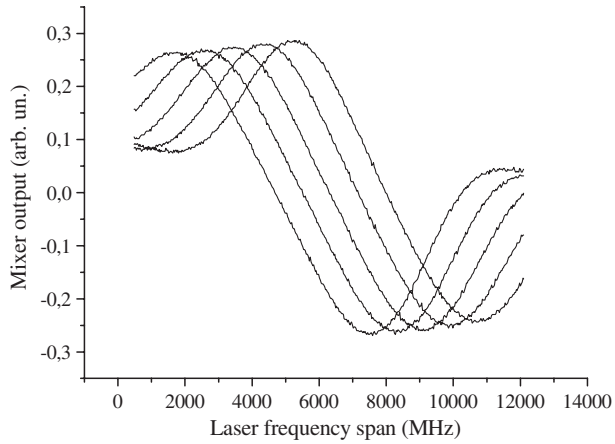

Figure 3. System response for increasing static strain along a laser frequency linear scan over the FBC centre. A series of equal 8 $\mu\varepsilon$ deformations were applied to the 50% FBG by a piezo actuator. The spectrum was recorded at a scan frequency of 13 Hz with a detection bandwidth of about 500 Hz.

3. Tests and results

A test for static strain measurements was performed on the 50% FBG. In particular, we investigated the response of the laser system for subsequent static loads on the FBG. Dispersive-like profiles were recorded with the laser frequency scanning while 8 $\mu\varepsilon$ voltage-equivalent steps were applied to the PZT fixed to the sensor. The resulting frequency shifts can be observed in figure 3. A frequency calibration of the DBM spectrum was obtained using a 1 GHz free-spectral-range Fabry–Perot interferometer. Moreover, since the curves generated at the DBM output have a direct amplitude-to-frequency correspondence, around the FBG centre, it is easy to relate amplitude variations to strain events without the need for centre frequency tracking. This means that the desired information is directly available at the DBM output even if the laser is not scanned around the FBG resonance. The system is able to perform strain measurements up to about 100 $\mu\varepsilon$ as a consequence of the linearity range of the FM signal. As can be seen in figure 3, the frequency dependence of the DBM output is quite linear within 4–5 GHz around the zero-crossing point. This aspect is strictly related to the intrinsic FBG spectral width, which can be basically increased using a higher-reflectivity element. On the other hand, an active feedback system that locks the laser frequency to the FBG peak would provide a wider dynamic operating range.

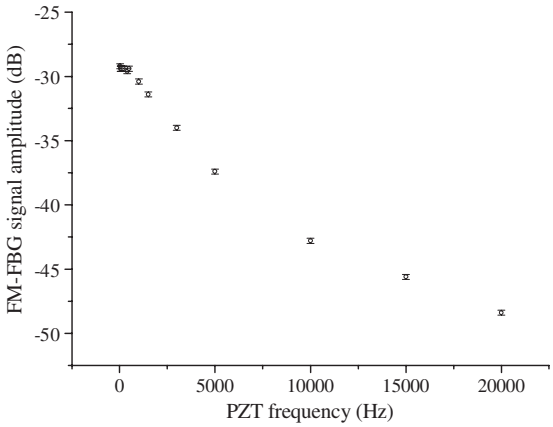

Figure 4. Relative amplitude response of the laser radio-frequency system to an acoustic excitation as a function of frequency. PZT capacitive behaviour becomes evident after 1 kHz. The vertical bars represent the maximum reading uncertainty of the amplitude values.

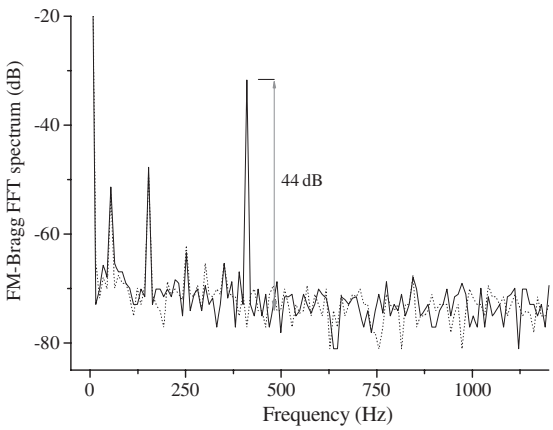

Figure 5. Dynamic response of the laser radio-frequency system to periodic deformations applied by the PZT with a strain-equivalent voltage of 4.5 $\mu\varepsilon$ at 410 kHz (resolution bandwidth 50 Hz). Background noise is still evident when the PZT is off in the 1–100 Hz range (dotted curve).

The measurement method showed a very fast dynamic response, as was pointed out in tests performed on the sensor in the acoustic domain. When an AC voltage was applied to the piezo actuator with the laser kept close to the resonance, the FFT spectrum of the DBM output exhibited an amplitude frequency response that went from DC up to 20 kHz, the highest value being determined by the piezo low-pass filtering (see figure 4). In particular, in figure 5, this is shown at 410 Hz. At this frequency, a 44 dB signal-to-noise ratio was observed, in a 50 Hz detection bandwidth, corresponding to a minimum detectable strain of 4 nε Hz$^{-1/2}$. A similar measurement was also performed at lower frequency, around 2 Hz and at 1 kHz, yielding detection limits of 150 nε Hz$^{-1/2}$ and 1.6 nε Hz$^{-1/2}$, respectively. To a first approximation, the sensitivity of our laser-based sensor may be likely to be limited by strain noise deriving from laser amplitude and frequency instabilities.

While thermally-induced refractive index fluctuations can be source of negligible phase errors, for a 1 cm-long FBG [19], shot-noise and laser-frequency noise may be of

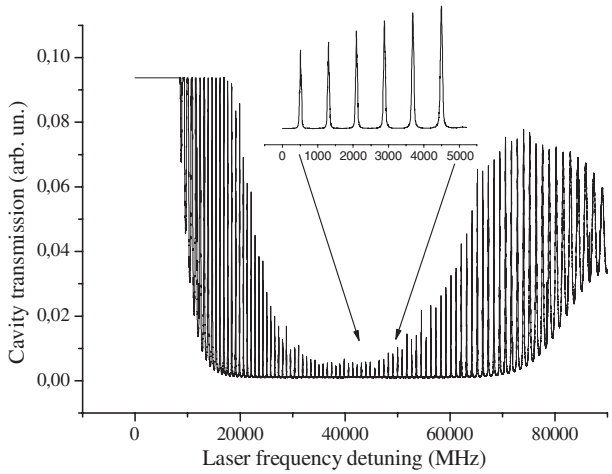

Figure 6. Transmission of the fibre resonator over the whole FBG bandwidth and on a magnified horizontal scale (inset).

some relevance. That can be deduced from the following expressions [8]

$$\varepsilon_{\text{shot-noise}} = \delta\nu/\delta V_{\text{FBG}} \frac{\sqrt{2eR_m R^2 G P B}}{0.8\nu} \tag{1}$$

$$\varepsilon_{\text{freq-noise}} = \frac{\sqrt{B\Delta f_{\text{L}}/\pi}}{0.8\nu}. \tag{2}$$

where $\delta\nu/\delta V_{\text{FBG}}$ is the linear slope of the FM dispersive signal in Hz V^{-1}, R_m is the maximum FBG reflectivity, R is the load resistance, G is the detector responsivity, B is the detection bandwidth, P is the incident power, Δf_{L} is the laser linewidth (reasonably assumed to be 20 MHz for a DFB laser) and ν is the optical frequency. The quantities in (1) and (2) yield a minimum detectable strain of 3×10^{-13} and 2×10^{-11} ε Hz$^{-1/2}$, respectively. Actually, a thorough investigation of laser frequency jitter would be necessary in order to extract a reliable estimate for (2). Also, laser frequency stability may seriously affect sensitivity and precision for strain measurements in the quasi-static domain and was thus tested through long-term reproducibility measurements.

For this purpose, a molecular ro-vibrational transition was used as a reference, and the line centre was continuously recorded over a time interval of about 4 h, measuring a root-mean-square deviation of about 4 MHz, i.e. 30 nε. Further limitations might be caused by optical feedback due to the residual reflectivity of fibre ends and optical components, giving rise to etalon effects and thus to power fluctuations at the detector. These kinds of fluctuations can be interpreted as false strain signals but are often difficult to evaluate. Furthermore, an unavoidable contribution may come from ambient acoustic noise.

Subsequently, we investigated the strain detection performance of the two-resonant-FBG array. In figure 6, its transmission, along a wide frequency interval, is shown. For this purpose, the laser diode was frequency-modulated by summing a large triangular-wave signal to the injection current. The graph exhibits an envelope shape due to the spectral overlap of the two FBGs, where the cavity finesse increases close to the FBG centre. A similar recording is also shown

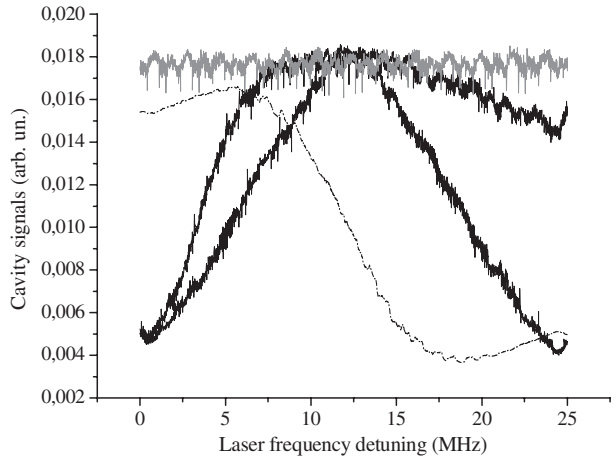

Figure 7. Transmission of the cavity and PDH error signal (dotted trace) during locking, for increasing servo gain.

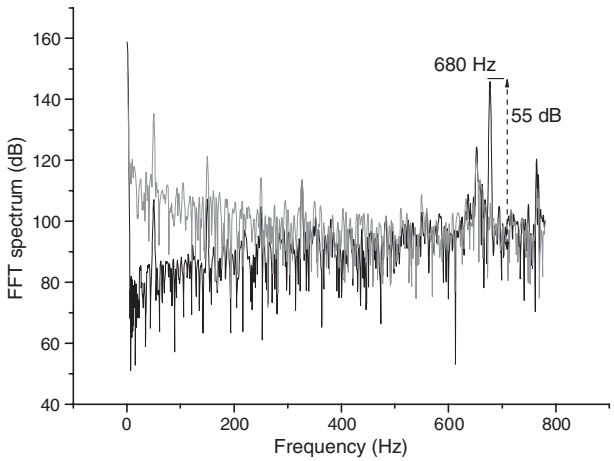

Figure 8. FFT spectrum of the PDH error signal when the laser is kept resonant with the cavity compared to the unlocked case in the upper trace (resolution bandwidth 2 Hz). Excitation of an acoustic wave of 127 nε amplitude at 680 Hz is detected.

in the inset of figure 6 for a narrower scan, which shows the Airy peaks in the centre region. In our scheme, the operational conditions have been chosen to satisfy a compromise between a high cavity finesse, deriving from a good overlap, and a reasonable transmission level (see inset of figure 6). In fact, the total intra-cavity transmission losses are significantly different among the excited longitudinal modes, as they depend on the equivalent optical pathlength and thus on the effective finesse (that is wavelength dependent). Furthermore, a fine structure with a splitting of a few tens of MHz can be observed in each cavity mode in the very high finesse region. This fact can be addressed to a weak birefringence into the photosensitive silica fibre adopted for the FBGs, leading to slightly nondegenerate resonances for different polarizations. For this reason, the polarization rotator is adjusted to select a single polarization state oscillating in the cavity. This phenomenon, as well as that caused by possible additional birefringence due to the FBG fabrication process, will be the subject of future investigations.

Using the PDH signal derived from the heterodyne detection system described above, the diode laser frequency is maintained locked to one cavity resonance. An example of the servo circuit operation is shown in figure 7 for increasing values of the closed-loop gain. The circuit corrects laser frequency fluctuations with respect to the cavity, directly acting on the injection current via the fast modulation input of the driver. Its action has been found to be effective over a time of several hours, in a bandwidth going from DC to about 200 Hz (unity-gain bandwidth). With this in mind, any mechanical or thermal stress applied to the cavity immediately translates to changes in error amplitude, thus providing a fast and sensitive strain monitor at frequencies beyond the servo unity-gain. Actually, perturbations acting within the servo bandwidth can still be observed in the PDH signal because of the imperfect loop correction. Nevertheless, the integrated error signal, fed to the laser input for corrections, can be used in this range for strain measurements as well. In figure 8, an example of strain detection with spectral amplitude analysis is shown. The lower trace is the FFT of the error signal in the tightly-locked case. In addition, an acoustic wave is excited in the sensing fibre by the PZT. The latter is driven by a 680 Hz AC

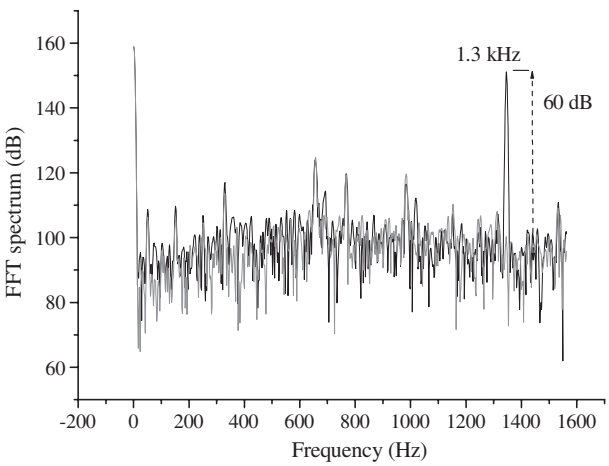

Figure 9. FFT spectrum of the error signal in the locked case with and without (grey trace) a dynamic strain at 1.3 kHz with a 40 nε amplitude (resolution bandwidth 5 Hz).

voltage with a strain-equivalent amplitude of 127 nε. Strain calibration can be carried out, as mentioned in the previous section, by sweeping the intra-cavity length over about 15% of the FSR while observing the cavity transmission with the laser sidebands. In this way, the sweep can be accurately frequency calibrated using the radio-frequency synthesizer and the corresponding length variation retrieved in terms of the PZT bias voltage as a function of frequency. As is visible in the FFT spectrum of figure 8, the action of the servo loop is clearly manifested in a maximum reduction of the error noise spectral density of nearly 50 dB, with respect to the free-running case (grey trace). At the same time, satellite oscillations are visible from 50 to 350 Hz, regardless of the servo action. These may be consequence of environmental noise that couples to the fibre. A first estimate of the signal-to-noise ratio at 680 Hz (45 dB), gives a minimum detectable strain of about 150 pε Hz$^{-1/2}$. This means that in the mid-frequency range, we achieved a sensitivity level improved by at least one order of magnitude with respect to the single-FBG

case. In figure 9, a similar plot is shown for higher-frequency operation, with a 1.3 kHz signal in the cavity. In this case, the resulting sensitivity is 20 pε Hz$^{-1/2}$.

A preliminary assessment of the potentialities of the system comes from the shot-noise-limited frequency noise spectral density of the PDH method [20]

$$S_{\mathrm{PDH}}(\mathrm{Hz\,Hz}^{-1/2}) = \frac{\Delta\nu_c}{J_0(\beta)}\sqrt{\frac{h\nu}{8\eta P_i}} \tag{3}$$

from which a minimum strain noise $\varepsilon_{\mathrm{PDH}} = S_{\mathrm{PDH}}/0.8\nu \cong 0.14$ fε can be derived, assuming the cavity bandwidth $\Delta\nu_c = 2$ MHz, the detector efficiency $\eta = 0.9$, the incident power $P_i = 0.1$ mW and the modulation depth $\beta = 0.3$. However, a more realistic approach would suggest considering free-running laser frequency fluctuations, which are not compensated by the locking system. Assuming that the laser is dominated by spontaneous-emission frequency noise, its emission spectrum would be Lorentzian with a linewidth corresponding to the previously specified value (20 MHz). Since laser frequency noise can be converted into cavity-length fluctuations, i.e. in strain noise, from (2) the ultimate strain limit corresponds to about 20 pε Hz$^{-1/2}$. The latter value is very close to the one we measured, at least for low-frequency servo operation.

4. Summary

We have demonstrated the possibility of using a spectroscopic radio-frequency modulation techniques for interrogation of FBGs and high-finesse fibre resonators made of two identical FBGs. A 1560 nm DFB diode laser was frequency modulated to generate sidebands at about 2.2 GHz. The sideband-to-carrier beat signal was shown to be extremely sensitive to Bragg wavelength shifts due to mechanical stress. Using this approach, both static and dynamic strain measurements could be performed, with a noise-equivalent sensitivity of the order of 150 nε Hz$^{-1/2}$ for slow deformations (2 Hz) and 1.6 nε Hz$^{-1/2}$ at higher acoustic frequencies (1 kHz). We showed the response of the whole system to acoustic vibration as a function of frequency in the range 0–20 kHz, demonstrating that this is essentially limited by the bandwidth of the test device. These issues are particularly relevant for geophysical monitoring applications of FBG sensors. Some aspects, related to sensitivity and precision in strain measurements, have also been studied to make the sensing apparatus suitable for outdoor use: our system is extremely compact, fast and versatile. An FBG sensor fabricated in a high-birefringence fibre will be considered in the future to provide a polarization-dependent sensitivity that allows discrimination between temperature and strain effects.

We also reported preliminary results on the use of the same laser in conjunction with a high-finesse Fabry–Perot cavity, made of two high-reflectivity FBGs. Frequency locking to the fibre cavity, by means of the PDH method with laser sidebands at about 11 MHz, was successfully carried out. The locking-loop feedback signals offered an efficient way of tracking mechanical deformations acting on the intra-cavity fibre length. A sensitivity gain of at least one order of magnitude, with respect to the single-FBG scheme, was demonstrated on the basis of preliminary tests, the set-up still being kept simple and compact. Work is currently in progress to further improve the performance of the locking system. Moreover, future efforts will be focused on the use of a narrower-linewidth laser source to push the sensitivity limits even further, aiming at the development of all-fibre-based acceleration, strain and temperature sensors for field use.

Acknowledgments

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